

Xinru Wang

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 [Google Scholar](#)

Summary My research intersects Human-Computer Interaction (HCI) and AI, with a focus on Explainable AI (XAI). I iterate on the **evaluation and design of user-centered XAI systems for AI-assisted human decision-making**. My research considers the contextual relevance of fast-evolving *AI models* and diverse *user interfaces*, with an emphasis on making AI explanations *collaborative, adaptable, and glanceable*.

Education **Purdue University** West Lafayette, IN
Ph.D. in Computer Science Aug. 2019 – Aug. 2025
Thesis: Human-Centered Evaluation and Design of AI Explanations in AI-Assisted Decision-Making
Thesis Committee: [Prof. Ming Yin](#) (Advisor), Prof. Ninghui Li, Prof. Alexandros Psomas, Prof. Tianyi Zhang

Peking University Beijing, China
B.S. in Intelligence Science and Technology Sep. 2015 – Jun. 2019
B.S. in Psychology (Dual Degree) Sep. 2016 – Jun. 2019

Professional Experience **Singapore-MIT Alliance for Research and Technology** Singapore
Postdoctoral Associate Sep. 2025 – Present
– Advisor: Prof. Thomas Malone

Adobe Research San Jose, CA, US
Research Intern May 2025 – Aug. 2025
– Mentor: Doga Dogan, Manager: Eunye Koh

Meta Reality Labs Redmond, WA, US
Research Intern Sep. 2023 – Mar. 2024
– Mentor: Anna Mengjie Yu, Manager: Tanya Jonker

Megagon Labs Mountain View, CA, US
Research Intern May 2023 – Aug. 2023
– Mentors: Hannah Kim, Zhengjie Miao

University of Michigan Ann Arbor, MI, US
Summer Research Intern Jul. 2018 – Sep. 2018
– Advisor: Prof. Lionel Robert

Kendall Square Capital Beijing, China
Machine Learning Intern Jan. 2019 – Apr. 2019

DiDi Beijing, China
Machine Learning Intern Sep. 2018 – Jan. 2019

Publications *Conference and Journal Papers* * indicates equal contribution; † indicates corresponding author

One AI Signal, Many Human Judgments: A Bayesian Cascade Analysis of AI-based Credibility Indicators in Online Information Spread
Zhuoran Lu, Weilong Wang, Yangyang Yu, **Xinru Wang**, Zhuoyan Li, Zhiwei Liu and Sophia Ananiadou
ACM Conference on Human-AI Complementarity and Alignment (HCOMP). 2026.

Graphionale: How Graph Visualizations of LLM Rationales Affect Human Decision Making

Xinru Wang*, Zhexuan Ma*, Ming Yin, Shuai Ma, Thomas W. Malone.

The ACM Symposium on User Interface Software and Technology (UIST). 2026.

Towards Human-AI Deliberation: Design and Evaluation of LLM-Empowered Deliberative AI for AI-Assisted Decision-Making

Shuai Ma, Qiaoyi Chen, **Xinru Wang**, Chengbo Zheng, Zhenhui Peng, Ming Yin, Xiaojuan Ma.

Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2025.
[\(Best Paper Honorable Mention\)](#)

Less or More: Towards Glanceable Explanations for LLM Recommendations Using Ultra-Small Devices

Xinru Wang, Mengjie Yu, Hannah Nguyen, Michael Iuzzolino, Tianyi Wang, Peiqi Tang, Natasha Lynova, Co Tran, Ting Zhang, Naveen Sendhilnathan, Hrvoje Benko, Haijun Xia, Tanya Jonker.

Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI). 2025.

Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels

Xinru Wang, Hannah Kim, Sajjadur Rahman, Kushan Mitra, Zhengjie Miao.

Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2024.

“Are You Really Sure?” Understanding the Effects of Human Self-Confidence Calibration in AI-Assisted Decision Making

Shuai Ma, **Xinru Wang**, Ying Lei, Chuhan Shi, Ming Yin, Xiaojuan Ma.

Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI). 2024.

The Effects of AI Biases and Explanations on Human Decision Fairness: A Case Study of Bidding in Rental Housing Markets

Xinru Wang, Chen Liang, Ming Yin.

Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI). 2023.

Who Should I Trust: AI or Myself? Leveraging Human and AI Correctness Likelihood to Promote Appropriate Trust in AI-Assisted Decision-Making

Shuai Ma, Ying Lei, **Xinru Wang**, Chengbo Zheng, Chuhan Shi, Ming Yin, Xiaojuan Ma.

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI). 2023.

Watch Out For Updates: Understanding the Effects of Model Explanation Updates in AI-Assisted Decision Making

Xinru Wang, Ming Yin.

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI). 2023.

Will You Accept the AI Recommendation? Predicting Human Behavior in AI-Assisted Decision Making

Xinru Wang*, Zhuoran Lu*, Ming Yin.

Proceedings of the 2022 ACM Web Conference (WWW). 2022.

Effects of Explanations in AI-Assisted Decision Making: Principles and Comparisons

Xinru Wang, Ming Yin.

ACM Transactions on Interactive Intelligent Systems (TiiS). 2022.

Are Explanations Helpful? A Comparative Study of the Effects of Explanations in AI-Assisted Decision-Making

Xinru Wang, Ming Yin.

Proceedings of the 26th International Conference on Intelligent User Interfaces (IUI). 2021.

Workshop Papers, Posters and Extended Abstracts

Demonstrating Graphionale: How Graph Visualizations of LLM Rationales Affect Human Decision Making

Zhexuan Ma*, Xinru Wang*, Ming Yin, Vasiliki Charisi, Shuo Sun, Alok Prakash, Shuai Ma, Thomas W. Malone.

The ACM Symposium on User Interface Software and Technology (UIST Demo). 2026.

Human-AI Collaborative Ontology Editing: An LLM-Assisted Dashboard for Improving an Ontology of Work Activities

Yilin Yang, Sophia Min, Alice Cai†, Xinru Wang†, Aadya Nishtha, Shuo Sun, Iman Yeckehzaare, Thomas Malone

ACM Collective Intelligence (CI Poster). 2026.

Where can AI be used? Insights from a deep ontology of work activities

Alice Cai, Iman YeckehZaare, Shuo Sun, Vasiliki Charisi, Xinru Wang, Aiman Imran, Robert Laubacher, Alok Prakash, Thomas W. Malone

ACM Collective Intelligence (CI Talk). 2026.

Persistent Signals, Absent Meaning: How Terminal-Stage Patients Navigate Health Data Exposure in Everyday Ubiquitous Computing Environments

Yibo Meng, Bingyi Liu, Xinru Wang, Zhuoyan Li, Zhuoran Lu

UbiComp/ISWC Poster. 2026.

The RoboAtlas: Mapping the Global Robotics Landscape

Jiacheng Zhang, Shuo Sun, Vicky Charisi, Xinru Wang, Xinyue Chen, Zhexuan Ma, Alok Prakash, Thomas Malone.

2026 IEEE International Conference on Robotics and Automation (ICRA Late-Breaking Result). 2026.

XAgen: An Explainability Tool for Identifying and Correcting Failures in Multi-Agent Workflows

Xinru Wang, Ming Yin, Eunye Koh, Mustafa Doga Dogan.

Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA). 2026

Human-Centered Evaluation of Explanations in AI-Assisted Decision-Making

Xinru Wang.

Companion Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI Doctoral Consortium). 2024.

Pre-prints and Under Review

From Citations to Evidence: Citation Provenance in Citation-supported AI Answers

Yuxuan Du, Phongsakon Mark Konrad, Shuai Ma, Qiao Jin, Xinru Wang, Zhuoran Lu

Calibrated Step-Level Confidence for Selective Human Oversight of AI Agents

Shuai Ma, Baiqiao Zhang, Q. Vera Liao, Zhuoran Lu, Xinru Wang, Jin Huang, Ming Yin, Qian Zhu, Feng Tian

From Review to Reuse: How Post-Task Workflow Can Support Human and AI Agent Interaction

Zekun Wu, **Xinru Wang**, Rock Pang, Chenglong Wang, Anna Feit

EmoCorrect: Scaling Early Intervention in Community Correction through AI-Driven Affective Monitoring

Kailai Sun*, **Xinru Wang***, Yue Cui, Guanyi Wang, Zijie Gao, Xinwei Wang, Qianchuan Zhao

From Agent Traces to Team Dynamics: A Relational Method for Interpreting Multi-Agent Behavior

Zhexuan Ma, Shuo Sun, Xinyue Chen, **Xinru Wang**, Vicky Charisi, Jiacheng Zhang, Alok Prakash, Thomas W. Malone

XAgent: Designing Explainability Support for Failure Detection and Correction in Multi-Agent Workflows

Xinru Wang, Zhuoran Lu, Shuai Ma, Ming Yin, Eunyee Koh, Mustafa Doga Dogan

When Ubiquitous Health Data Stops Helping: Patients' Experience of Breakdowns in the Data-Interpretation-Action Loop at the End of Life

Yibo Meng, **Xinru Wang**, Zhuoyan Li, Chengxi Zang, Zhuoran Lu, Zhicong Lu

Where can AI be used? Insights from a deep ontology of work activities

Alice Cai, Iman YeckehZaare, Shuo Sun, Vasiliki Charisi, **Xinru Wang**, Aiman Imran, Robert Laubacher, Alok Prakash, Thomas W. Malone

Beyond Recommender: An Exploratory Study of the Effects of Different AI Roles in AI-Assisted Decision Making

Shuai Ma, Chenyi Zhang, **Xinru Wang**, Xiaojuan Ma, Ming Yin.

Patent

Multi-AI Agent Explainability to Trace Dynamic Interactions Between AI Agents and AI Agent Tasks

Xinru Wang, Mustafa Doga Dogan, Eunyee Koh. *U.S. Patent Application (Adobe Reference No. P14400-US) (filed). 2025.*

A kind of menu based on deep learning recommends method

Xinru Wang, Yichun Wang, Zhilou Yu. *Chinese Patent CN107665254A (published). 2018.*

Selected
Talks

User-Centered Evaluation and Design for Human-AI Decision-Making

Presented at Zhejiang University

Apr. 2026

User-Centered Evaluation and Design of AI Explanations in AI-Assisted Decision-Making

Presented at the University of North Carolina at Chapel Hill

Feb. 2025

Understanding Human Behavior in AI-Assisted Decision Making: Experiments and Models

Presented at [MIDAS Future Leaders Summit 2022](#)

Apr. 2022

Service

Editorial Board

Computer-Human Interaction and Intelligence

Area Chair

ACM Conference on Designing Interactive Systems (DIS): 2026

ACM Conference on Human Factors in Computing Systems (CHI): 2027, 2026

Session Chair

ACM Conference on Human Factors in Computing Systems (CHI): 2026 (Explaining and Evaluating AI Systems)

Program Committee

ACM Conference on Fairness, Accountability, and Transparency (FAccT): 2025, 2023

ACM Symposium on Eye Tracking Research & Applications (ETRA): 2025

ACM Conference on Intelligent User Interfaces (IUI): 2025

ACM CHI Workshop on Human-Centered Explainable AI (HCXAI): 2024, 2023, 2022

Assistant to Subcommittee Chair

ACM Conference on Human Factors in Computing Systems (CHI): 2025

Conference Reviewer

ACM Conference on Human-AI Complementarity and Alignment (HCOMP): 2026

ACM Conference on Human Factors in Computing Systems (CHI) Poster: 2026

ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW): 2026

ACM Symposium on User Interface Software and Technology (UIST): 2026, 2025

ACM Conference on Human Factors in Computing Systems (CHI): 2025, 2024, 2023

ACM Conference on Human Factors in Computing Systems (CHI) Late-Breaking Work: 2025, 2024, 2023

ACM Conference on Designing Interactive Systems (DIS): 2024

ACM/IEEE International Conference on Human-Robot Interaction (HRI) alt.HRI: 2024

Journal Reviewer

Transactions on Computer-Human Interaction (TOCHI)

IEEE Transactions on Human-Machine Systems (THMS)

ACM Transactions on Interactive Intelligent Systems (TiiS)

Transactions on Intelligent Systems and Technology (TIST)

Student Volunteer

ACM Conference on Intelligent User Interfaces (IUI): 2024

ACM Conference on Research and Development in Information Retrieval (SIGIR): 2018

Teaching

CS475 Human-Computer Interaction, Purdue University

Graduate Teaching Assistant for Prof. Ming Yin

Spring 2025

CS471 Introduction to Artificial Intelligence, Purdue University

Head Graduate Teaching Assistant for Prof. Raymond A. Yeh

Spring 2023

Head Graduate Teaching Assistant for Prof. Ming Yin

Summer 2022, Fall 2022

Graduate Teaching Assistant for Prof. Jennifer Neville

Fall 2019

CS242 Introduction to Data Science, Purdue University

Graduate Teaching Assistant for Prof. Tony Bergstrom

Spring 2020

CS38001 Python Programming, Purdue University

Graduate Teaching Assistant for Prof. Ruby Tahboub

Fall 2019, Spring 2020

Mentoring

Yilin Yang (undergraduate student, Wellesley College)

Summer 2026

Sophia Min (undergraduate student, MIT)

Summer 2026

Xinyue Chen (M.S. student, National University of Singapore)

Spring 2026

Zhexuan Ma (M.S. student, National University of Singapore)

Spring 2026

Songyu He (undergraduate student, HKUST)

Fall 2024

Hannah Nguyen (research assistant, Meta Reality Labs)

Spring 2024

Honors and Awards	Best Paper Honorable Mention (CHI 2025)	2025
	NSF Student Travel Award (IUI 2024)	2024
	Apple PhD Fellowship Nominee (Purdue University)	2024
	Graduate Teaching Award (Purdue University)	2023
	Summit Attendee and Speaker (MIDAS Future Leaders Summit)	2022
	Women in Science Program Travel Grant (Purdue University)	2022
	Microsoft Research PhD Fellowship Nominee (Purdue University)	2022
	Academic Excellence Award (Peking University)	2018
	Fei-Xun Scholarship (Peking University)	2018
	Academic Excellence Award (Peking University)	2016
	May 4th Scholarship (Peking University)	2016